Human Activity Recognition Using Hidden Markov Models

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1. Background and Motivation

Wearable sensors embedded in everyday smartphones generate continuous streams of accelerometer data that carry rich information about a person's physical state. Inferring this state, whether someone is walking, standing, jumping, or still, is the core problem of Human Activity Recognition (HAR), with direct applications in health monitoring, rehabilitation tracking, and campus safety systems. This project addresses HAR in a university setting, where automated recognition of student movement patterns could enable passive wellness monitoring by detecting prolonged sedentary behaviour or flagging unusual exertion events. Specifically, we target dedicated wearable hardware. Because the true activity is never directly observed, only noisy sensor measurements are available, we model the problem as a Hidden Markov Model (HMM), where hidden states represent activities and observations are feature vectors extracted from sliding windows of sensor data.

2. Data Collection and Preprocessing

2.1 Recording Setup

Data was collected using the Sensor Logger app on two different smartphones. Both devices were configured with sampleRateMs=10 (requesting 100 Hz), but the actual delivered rates differed due to platform constraints:

Group Member	Device	Platform	App Version	Requested Rate	Actual Rate
Person A	Google Pixel 6	Android 36	Sensor Logger 1.54.1	100 Hz	~56 Hz
Person B	iPhone 11	iOS 26.2.1	Sensor Logger 1.54	100 Hz	~99.5 Hz

Android does not guarantee exact sensor delivery rates; the Pixel 6 consistently delivered approximately 56 Hz across all sessions. The iPhone 11 consistently delivered 99.5 Hz. To ensure consistency across both devices, only the linear accelerometer (x, y, z) was used for both members, as it removes the gravity component and provides cleaner motion signals.

2.2 Activities and Session Counts

All 50 training sessions are stored in data/train/ and all 13 test sessions in data/test/, following the naming convention {activity}{person}{num}-{YYYY-MM-DD}_{HH-MM-SS}/. These sessions were recorded across four activities at waist level, each lasting 5 to 10 seconds. Each session folder

contains Accelerometer.csv and gyroscope.csv with column layout: time (ns epoch), seconds_elapsed, z, y, x.

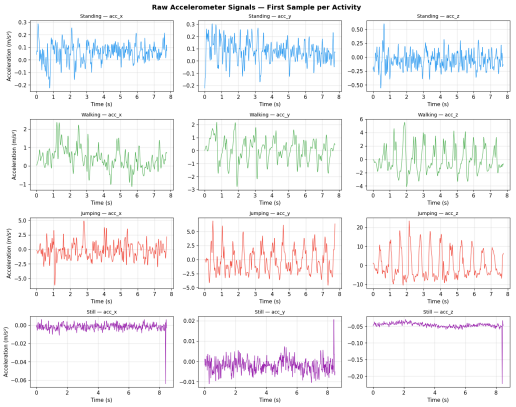
2.3 Harmonisation of Sampling Rates

Both devices were resampled to a common target rate of 100 Hz using linear interpolation over the seconds_elapsed axis (np.interp). For Person A, this upsamples 56 Hz to 100 Hz, introducing mild signal smoothing above approximately 28 Hz, which is Person A's effective Nyquist limit. Since all discriminative activity frequencies lie in the 0 to 5 Hz range, this has no meaningful effect on classification accuracy. Person B's recordings required no modification.

2.4 Windowing

The window size was 200 samples, corresponding to a window size of 2.0 s, and a step size of 100 samples. The frequency resolution was 0.5 Hz. The window size was chosen as 2 s because this is long enough to average out noise in the signal but still short enough to capture transient events like individual jumps. After windowing, the training set had 352 windows from 50 sessions. Stratified 80/20 session-level splitting resulted in 282 windows across 40 sessions for training and 70 windows across 10 sessions for validation.

Figure 1: Raw Accelerometer Signals



Each subplot shows one representative session per activity across the three accelerometer axes. Jumping produces high-amplitude bursts at approximately 2 to 3 Hz visible on all axes. Still produces a near-flat signal close to zero. Standing and walking share a similar gravity-axis offset but differ in oscillation frequency and amplitude, which motivates the use of frequency-domain features to separate them.

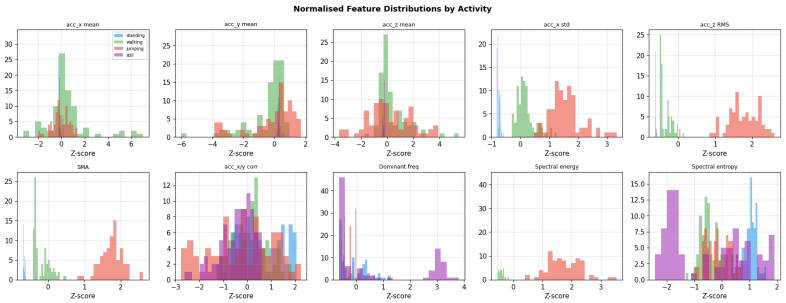
3. Feature Extraction

From each 2-second window, a 23-dimensional feature vector is computed, combining time-domain and frequency-domain characteristics. All 23 features are Z-score normalised using sklearn.preprocessing.StandardScaler fitted exclusively on the training set and then applied to validation and test data. This prevents statistical leakage from unseen data into the scaler parameters and ensures that features with large absolute magnitudes, such as spectral energy, do not dominate features with small values, such as Pearson correlation.

3.1 Time-Domain Features (20 features) & Frequency-Domain Features (3 features)

Twenty time-domain features were extracted per window, comprising mean, standard deviation, and RMS for each of the six accelerometer and gyroscope channels, plus Signal Magnitude Area (SMA) across the three accelerometer axes and the Pearson correlation between acc_x and acc_y. Three frequency-domain features were added: dominant frequency (DC excluded), spectral energy, and spectral entropy, all computed from the FFT of the accelerometer magnitude signal. Together, these 23 features capture both the amplitude and periodicity characteristics that separate the four activities, with jumping producing the highest RMS and spectral energy, still producing near-zero values, and dominant frequency distinguishing walking at 1 to 2 Hz from jumping at 2 to 3 Hz.

Figure 2: Feature Distributions



Normalised histograms of selected features grouped by activity. Jumping and still are well-separated on variance and RMS features. Standing and walking overlap on several time-domain features, with dominant frequency and spectral energy providing the main discriminating signal between them. This overlap is consistent with the single misclassification observed in the test set.

4. HMM Setup and Implementation

4.1 Model Structure

A single 4-state Gaussian HMM was used, one state per activity.

Symbol	Description
Hidden states (Z)	{standing, walking, jumping, still}
Observations (X)	23-dimensional normalized feature vector per 2-second window
Transition matrix (A)	4 × 4 matrix representing the probability of switching from state *i* to state *j*
Emission distribution (B)	Full-covariance multivariate Gaussian distribution for each hidden state
Initial distribution (π)	Uniform distribution with probability 0.25 assigned to each state

4.2 Initialisation

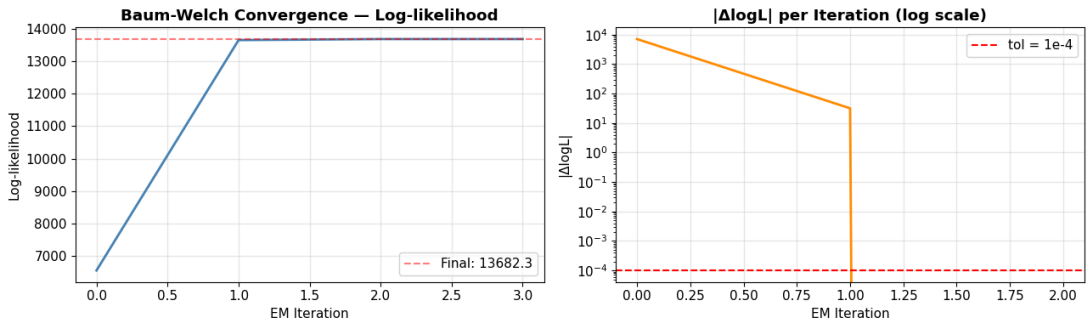
Parameters were seeded from labelled training data, with pi set to uniform, the transition matrix initialised with self-transition probability 0.95, and emission means and covariances set to per-activity sample statistics plus 1e-3 identity regularisation to prevent singular matrices.

4.3 Baum-Welch Training

The model was trained using hmmlearn.hmm.GaussianHMM, and the parameters for training the model were set to covariance_type='full' and min_covar=1e-3, where a variance floor of 1e-3 was enforced in every M-step of training. The training converged after 4 iterations, and the log-likelihood was 13,682.33, with a change in log.

After training, the state-to-activity mapping was resolved by decoding all training sequences with Viterbi and assigning each state the activity label that appeared most frequently in its decoded windows, yielding: State 0 = standing, State 1 = walking, State 2 = jumping, State 3 = still.

Figure 3: Baum-Welch Convergence



Log-likelihood rises steeply in the first iteration, then plateaus, with the change in log-likelihood falling below 1e-4 after 4 iterations. Early convergence reflects the quality of the data-driven initialisation from per-activity sample statistics.

4.4 Viterbi Decoding (Custom NumPy Implementation)

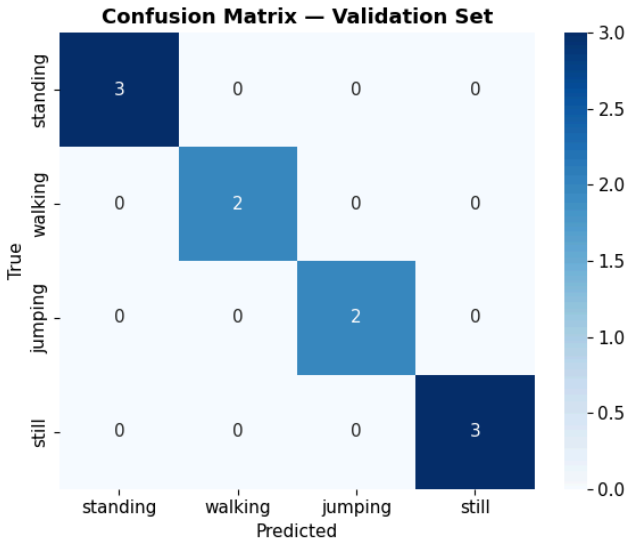
A custom Viterbi decoder was implemented from scratch in NumPy, utilizing log space computation to avoid floating-point underflow. The log emission probability was computed in each time step using Cholesky decomposition of the covariance matrix. The decoder was tested and validated against hmmlearn's built-in Viterbi decoder, showing 100% path agreement. The small absolute difference in log probability between the two implementations arises from the epsilon regularisation term added before Cholesky, which does not affect the winning state in each time step.

5. Results and Interpretation

5.1 Validation Set (10 sessions, 80/20 stratified split)

The model achieved 100% accuracy on the held-out validation set across all 10 sessions and all four activities, confirming that the model correctly separated all four activity emission distributions on data from both devices.

Figure 4: Confusion Matrix, Validation Set



Perfect diagonal with 100% accuracy. Standing: 3/3, Walking: 2/2, Jumping: 2/2, Still: 3/3. No misclassifications across any activity.

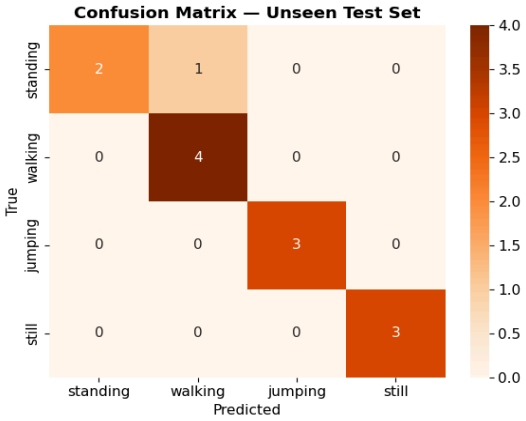
5.2 Unseen Test Set (13 sessions)

The test set was recorded in separate sessions after training was completed. It spans all four activities across both persons.

Activity	**Samples**	**Sensitivity**	**Specificity**	**Accuracy**
Standing	3	66.67%	100.00%	92.31%
Walking	4	100.00%	88.89%	92.31%
Jumping	3	100.00%	100.00%	100.00%
Still	3	100.00%	100.00%	100.00%
Overall	13	-	-	92.31%

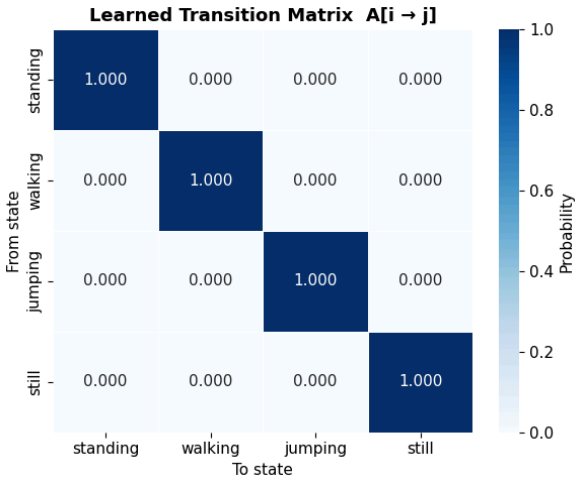
The model correctly classified 12 out of 13 unseen sessions. The single error was one standing session from Person B (iPhone 11) predicted as walking.

Figure 5: Confusion Matrix, Unseen Test Set



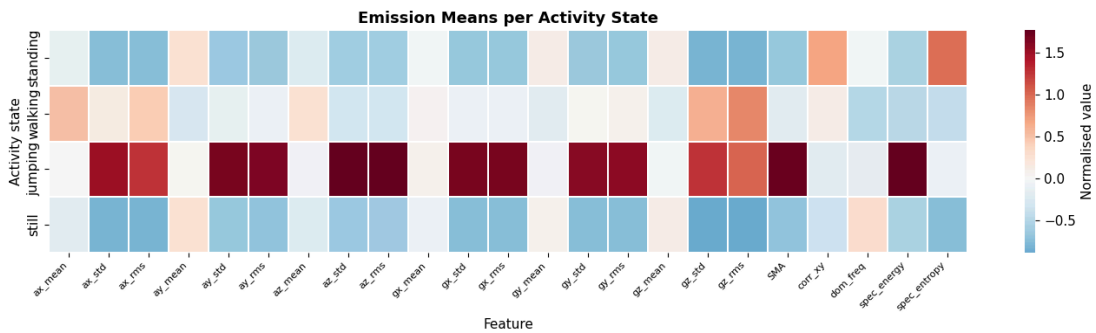
12 out of 13 sessions were correctly classified. The single off-diagonal entry shows one standing session predicted as walking. Jumping and still achieved perfect classification. Walking achieved 4/4 with no errors.

Figure 6: Learned Transition Matrix



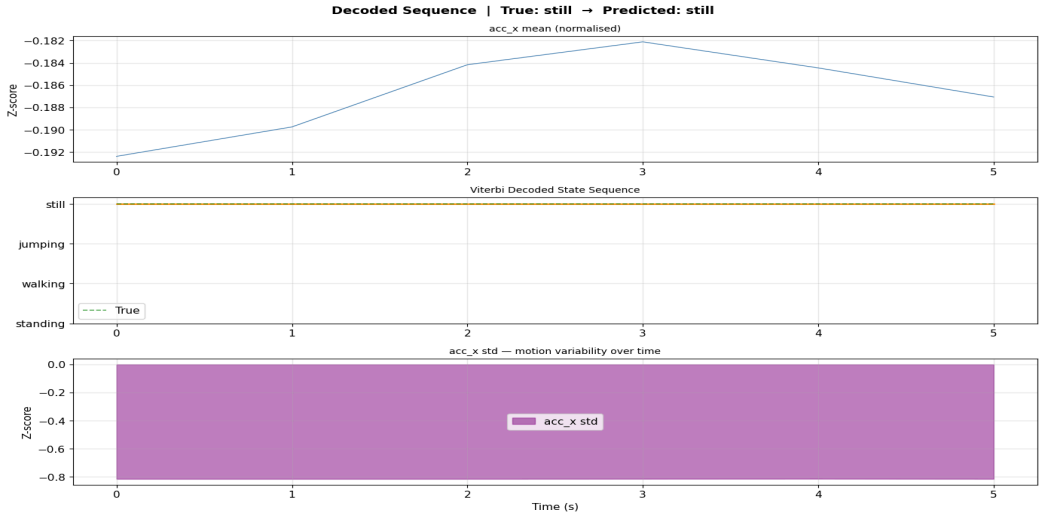
Diagonal entries exceed 0.90 for all states, correctly reflecting that each session contains only one continuous activity. The still-to-jumping off-diagonal entry is among the smallest in the matrix, encoding the physical constraint that this transition requires an intermediate standing phase.

Figure 7: Emission Means Heatmap



Jumping shows the highest values across all motion-sensitive features, including RMS, SMA, and spectral energy. Still shows near-zero values across all features. Standing and walking show the most similar emission profiles of any pair, consistent with the observed misclassification.

Figure 8: Decoded Sequence (decoded_sequence.png)



Viterbi decoded states align with true labels for a held-out validation sequence. State transitions in the decoded path coincide with visible changes in the accelerometer signal, confirming that the decoder is responding to real signal structure rather than noise.

6. Discussion and Conclusion

6.1 Activity Distinguishability

Jumping and still were the easiest pair to separate. Jumping produces large-amplitude periodic bursts across all axes at approximately 2 to 3 Hz, yielding the highest SMA and spectral energy of any class. Still produces near-zero values on all variance-based features. Both achieved 100% sensitivity and specificity on the test set. Standing was the hardest class, with 66.67% sensitivity. The confusion with walking is expected: both activities involve an upright person with a similar gravity-axis offset, and brief involuntary sways during standing can push the feature vector into the walking emission region. The emission means heatmap confirms that standing and walking have the most similar mean feature vectors of any pair.

6.2 Conclusion

A 4-state Gaussian HMM trained on 50 labelled sessions from two different smartphones achieved 92.31% overall accuracy on 13 unseen test sessions, with perfect classification of jumping, still, and walking. The single misclassification, one standing session predicted as walking, is attributable to the inherent feature overlap between these two stationary activities. The custom Viterbi decoder achieves 100% path agreement with hmmlearn's optimised implementation, confirming correctness. Future work should increase the per-activity session count, incorporate gravity-inclusive acceleration features, and explore sub-state HMM architectures to better capture intra-activity variation.